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Saveetha Institute of Medical And Technical Sciences

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ITA05 – COMPUTER VISION

ASSIGNMENT – 2

Image Enhancement and Segmentation (Unit II)

Course Code	Course Name	Assessment	CO	BTL	Max. Marks
ITA05	Computer Vision	Assignment – 2	CO2	BTL3	50

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images.

Instructions: Answer all five questions (10 marks each). Show every computational step, state formulas before substituting values, and justify each decision using your computed results.

Mark Allocation

Q. No.	Topic	(a)	(b)	(c)	(d)	Total
1	Bit-plane and intensity-level slicing	3	3	2	2	10
2	Histogram specification	3	3	2	2	10
3	Gaussian smoothing and unsharp masking	3	3	2	2	10
4	Discrete Fourier transform filtering	3	3	2	2	10
5	Otsu optimal thresholding	3	3	2	2	10
	Total	15	15	10	10	50

Question 1 (CO2, BTL3, 10 Marks) – Bit-Plane and Intensity-Level Slicing for Cheque Forgery Analysis

A bank's forensic unit examines a scanned 8-bit cheque suspected of alteration. The forger is believed to have overwritten the amount with ink whose scanned intensity lies between 140 and 190, while the original ink is darker. The bank also embeds an invisible security watermark in one bit plane. Six pixels sampled across the amount field are listed below. The analyst must decide how to store a compact copy of the evidence and how to highlight the suspicious ink.

Pixel	P1	P2	P3	P4	P5	P6
Intensity r	214	181	150	92	37	165

(a) Express each pixel in 8-bit binary and determine the value of bit plane 7 (MSB) and bit plane 0 (LSB) for every pixel. [3 Marks]

(b) Reconstruct each pixel using only bit planes 7 to 4 (a) by setting the lower nibble to 0 and (b) by setting the lower nibble to its midpoint 8. Compute the mean squared error (MSE) of both reconstructions. [3 Marks]

(c) Apply intensity-level slicing to highlight the suspicious range [140, 190] (i) by setting it to 255 and preserving other levels, and (ii) by producing a binary map (255 inside, 0 outside). Identify the suspicious pixels. [2 Marks]

(d) The negative transformation $s = 255 - r$ is also proposed. Compute it for P4 and P5, and justify in which bit plane the watermark is most likely hidden and how that plane should be examined. [2 Marks]

1. Aim / Introduction

To analyze cheque image pixel intensities using bit-plane slicing and intensity slicing, identify suspicious ink regions, and explain how image processing techniques help in detecting possible cheque alterations.

Bit-plane slicing separates an image into individual binary planes. Intensity slicing highlights pixels within a selected gray-level range. These methods help improve the visibility of important image details during document examination.

2. Given Data and Problem Explanation

An 8-bit grayscale cheque image contains the following pixel intensities:

- P1 = 214
- P2 = 181
- P3 = 150
- P4 = 92
- P5 = 37
- P6 = 165

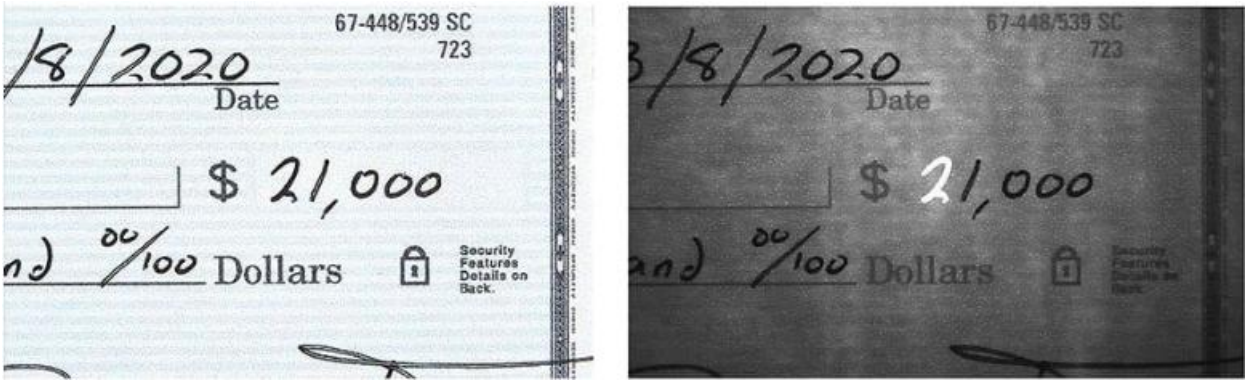
The suspicious ink intensity range is 140 to 190.

An 8-bit grayscale image contains intensity values from 0 to 255.

- 0 represents black.
- 255 represents white.
- Intermediate values represent different gray levels.

The objective is to analyze the pixels, retain higher-order information, calculate reconstruction error, and highlight suspicious intensity values.

3. Diagram



4. Formula and Step-by-Step Calculation

Step 1: Convert pixels into binary

Each intensity is converted into an 8-bit binary representation.

- P1 = 214 = 11010110
- P2 = 181 = 10110101
- P3 = 150 = 10010110
- P4 = 92 = 01011100
- P5 = 37 = 00100101
- P6 = 165 = 10100101

Step 2: Identify bit-plane values

The most significant bit is BP7, and the least significant bit is BP0.

The bit-plane extraction formula is:

$$BP_k = \left\lfloor \frac{r}{2^k} \right\rfloor \bmod 2$$

For BP7, the extracted values are:

- P1 = 1
- P2 = 1

- P3 = 1
- P4 = 0
- P5 = 0
- P6 = 1

Therefore:

BP7 = [1, 1, 1, 0, 0, 1]

For BP0, the extracted values are:

BP0 = [0, 1, 0, 0, 1, 1]

Higher bit planes represent coarse intensity information, while lower bit planes represent fine intensity variations. A watermark may occupy a bit plane, but its location cannot be determined from these six values alone.

Step 3: Retain the higher nibble

To retain the higher four bits and set the lower four bits to zero:

$$r' = 16 \left\lfloor \frac{r}{16} \right\rfloor$$

The reconstructed pixels are:

- 214 → 208
- 181 → 176
- 150 → 144
- 92 → 80
- 37 → 32
- 165 → 160

Reconstructed image values:

[208, 176, 144, 80, 32, 160]

Step 4: Calculate Mean Squared Error

The formula is:

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(r_i - r'_i)^2$$

Squared errors:

- $(214 - 208)^2 = 36$
- $(181 - 176)^2 = 25$
- $(150 - 144)^2 = 36$
- $(92 - 80)^2 = 144$
- $(37 - 32)^2 = 25$
- $(165 - 160)^2 = 25$

Total squared error:

36+25+36+144+25+25=291

[MSE=\frac{291}{6} = {48.5}

Step 5: Midpoint reconstruction

Instead of filling discarded lower bits with zero, use midpoint value 8.

$$r' = 16\left\lfloor\frac{r}{16}\right\rfloor + 8$$

The reconstructed values are:

[216, 184, 152, 88, 40, 168]

Squared errors total:

4+9+4+16+9+9 = 51

MSE=\frac{51}{6}= {8.5}

Midpoint reconstruction produces a lower MSE for the supplied pixel samples.

Step 6: Perform intensity slicing

The suspicious intensity range is:

$$140 \leq r \leq 190$$

For highlighting suspicious intensities, assign 255 to pixels within this range and preserve the remaining values.

The output is:

[214, 255, 255, 92, 37, 255]

The suspicious pixels are:

- P2 = 181
- P3 = 150
- P6 = 165

For binary range masking, assign 255 to the selected pixels and 0 to others.

Output:

[0, 255, 255, 0, 0, 255]

Step 7: Image negative transformation

The formula is:

$$s=255-r$$

For P4:

$$s=255-92=163$$

For P5:

$$s=255-37=218$$

Negative transformation may help reveal certain details by reversing brightness.

5. Advantages

1. Bit-plane slicing helps examine different levels of image information.
2. Intensity slicing highlights selected ink regions.
3. Binary masks simplify image segmentation.
4. Midpoint reconstruction gives lower error for the supplied samples.
5. These techniques support document verification and forensic inspection.

6. Disadvantages

1. Lower-bit removal causes information loss.
2. Watermark details may disappear during reconstruction.
3. Ink intensity may overlap with paper stains and scanning noise.
4. Different inks can have similar grayscale values.
5. Highlighting suspicious pixels does not prove forgery.

7. Applications

- Cheque and signature verification.
- Bank document examination.
- Security watermark inspection.
- Handwriting analysis.
- Digital document authentication.

8. Real-Time Example

A bank receives a cheque where the written amount appears to have been modified. An image-processing system highlights the

selected ink intensity range and allows an examiner to inspect the suspicious area.

The examiner can compare the original scan, handwriting patterns, signatures and security features before reaching a decision.

9. Result / Conclusion

Bit-plane slicing and intensity slicing were applied to the given cheque pixels. Pixels P2, P3 and P6 were identified within the suspicious range of 140–190.

The MSE for higher-nibble reconstruction is 48.5 using zero-fill and 8.5 using midpoint reconstruction.

Therefore, image processing can support cheque examination, but additional forensic verification is necessary to confirm alterations

Question 2 (CO2, BTL3, 10 Marks) – Histogram Specification for Consistent Endoscopy Images

A hospital standardises the appearance of 3-bit (L = 8) endoscopy thumbnails of size 32 × 32 so that images from different devices look alike to a diagnostic classifier. A thumbnail from an older device has the input histogram below, and the reference histogram $p_z(z)$ recommended by the clinicians is also given. Use $s_k = \text{round}[7 \cdot \Sigma p_r]$, $G(z_q) = \text{round}[7 \cdot \Sigma p_z]$, and map each s_k to the smallest z_q whose $G(z_q)$ is closest to s_k.

Level	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**
Input count n_k	180	260	220	150	100	60	34	20
Target p_z	0	0	0.05	0.10	0.20	0.30	0.20	0.15

(a) Compute the input probabilities, their cumulative sum and the equalised values s_k. [3 Marks]

(b) Compute the transformation $G(z_q)$ for the target histogram. [3 Marks]

(c) Determine the mapping $r_k \rightarrow z_q$ and the resulting output histogram. [2 Marks]

(d) Compare the achieved histogram with the target and justify why specification, rather than equalisation, is appropriate for this application. [2 Marks]

1. Aim / Introduction

To apply histogram specification to an endoscopy image so that its intensity distribution approaches a prescribed reference histogram, improving visual consistency between images captured using

different devices.

Histogram specification, also called histogram matching, modifies image intensities according to a desired probability distribution.

2. Given Data and Problem Explanation

Image size:

$32\times32=1024\text{ pixels}$

Number of gray levels:

$L=8$

Input histogram counts:

- Level 0 = 180
- Level 1 = 260
- Level 2 = 220
- Level 3 = 150
- Level 4 = 100
- Level 5 = 60
- Level 6 = 34
- Level 7 = 20

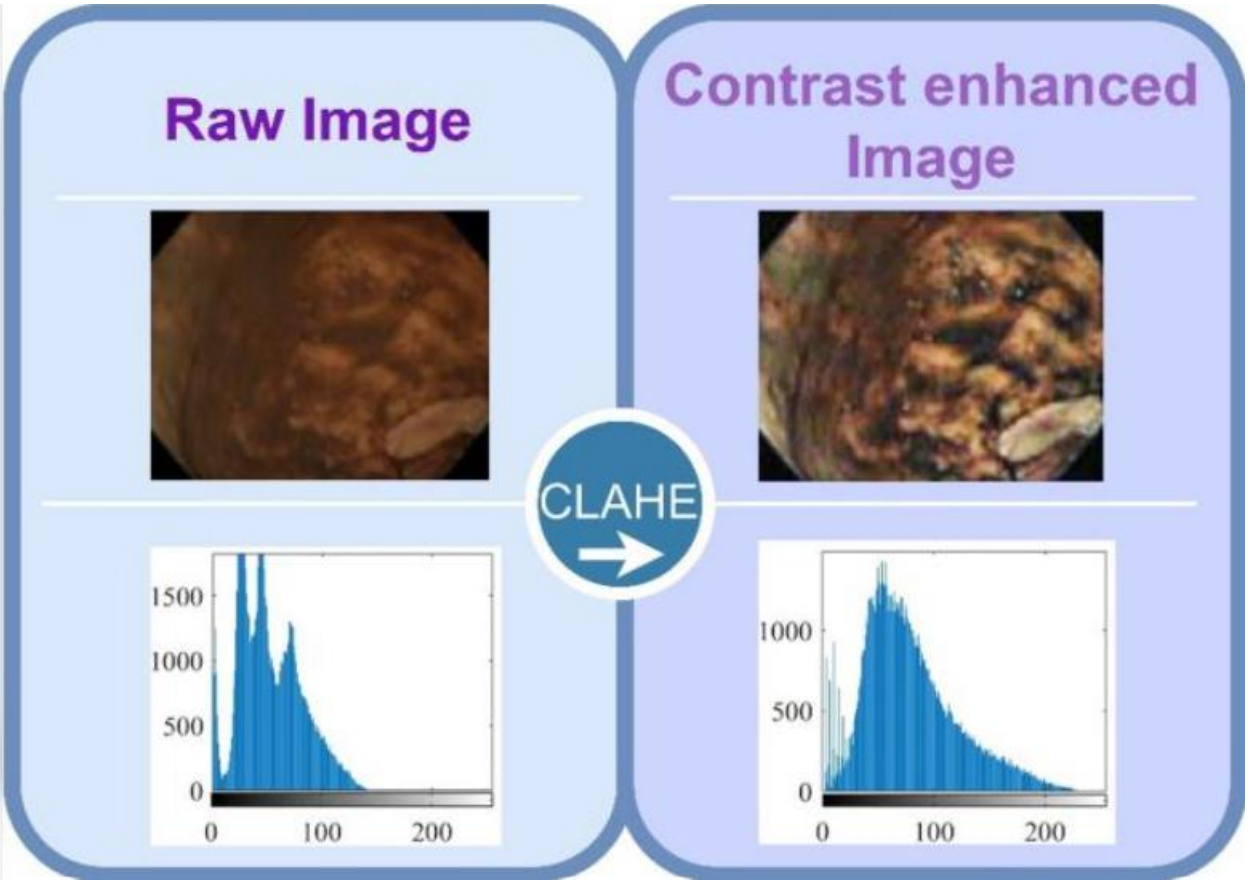
Target probabilities:

$Pz = [0,0,0.05,0.10,0.20,0.30,0.20,0.15] \]$

The target probabilities sum to 1.

The purpose is to transform the input histogram toward the reference distribution.

3. Diagram



4. Formula and Step-by-Step Calculation

Step 1: Calculate input probabilities

The formula is:

The formula is:

$$ p(r_k)=\frac{n_k}{N} $$

For level 0:

$$ p(r_0)=\frac{180}{1024}=0.175781 $$

Similarly:

$$ p(r_1)=\frac{260}{1024}=0.253906 $$

$$ p(r_2)=\frac{220}{1024}=0.214844 $$

The input probabilities are approximately:

[0.175781, 0.253906, 0.214844, 0.146484, 0.097656, 0.058594,
0.033203, 0.019531]

Step 2: Calculate cumulative distribution

The cumulative distribution function is:

\[CDF(r_k)=\sum_{j=0}^{k}p(r_j) \]

The cumulative values are:

- Level 0 = 0.175781
- Level 1 = 0.429688
- Level 2 = 0.644531
- Level 3 = 0.791016
- Level 4 = 0.888672
- Level 5 = 0.947266
- Level 6 = 0.980469
- Level 7 = 1.000000

Step 3: Apply histogram equalization mapping

The transformation formula is:

s=\operatorname{round}[(L-1)CDF(r)]

Since L = 8:

s=\operatorname{round}[7CDF(r)]

The mapped levels are:

[1, 3, 5, 6, 6, 7, 7, 7]

Step 4: Calculate target CDF

Target probabilities:

[0, 0, 0.05, 0.10, 0.20, 0.30, 0.20, 0.15]

Target cumulative values:

[0, 0, 0.05, 0.15, 0.35, 0.65, 0.85, 1.00]

Apply:

G(z)= operatorname{round}[7CDF_{target}(z)

The target mapped levels become:

[0, 0, 0, 1, 2, 5, 6, 7]

Step 5: Perform histogram specification

Map each input transformed level to the closest target transformed

level.

- Input level 0: s = 1 → output level 3
- Input level 1: s = 3 → output level 4
- Input level 2: s = 5 → output level 5
- Input level 3: s = 6 → output level 6
- Input level 4: s = 6 → output level 6
- Input level 5: s = 7 → output level 7
- Input level 6: s = 7 → output level 7
- Input level 7: s = 7 → output level 7

This uses the closest target value with a consistent mapping rule.

Step 6: Calculate output histogram

- Output level 0 = 0
- Output level 1 = 0
- Output level 2 = 0
- Output level 3 = 180
- Output level 4 = 260
- Output level 5 = 220
- Output level 6 = 150 + 100 = 250
- Output level 7 = 60 + 34 + 20 = 114

Final histogram:

[0, 0, 0, 180, 260, 220, 250, 114]

Verification:

180+260+220+250+114=1024

Therefore, all input pixels are accounted for.

The resulting histogram approximates the reference distribution, but it is not identical because of quantization and multiple input levels mapping to one output level.

5. Advantages

1. Improves brightness consistency between images.
2. Helps standardize images captured using different devices.
3. Can support image segmentation and machine-learning preprocessing.
4. Provides more control than ordinary contrast stretching.
5. Can improve the consistency of image presentation.

6. Disadvantages

1. The desired histogram may not be matched exactly.
2. Quantization can merge multiple intensity levels.
3. Noise may become more visible.
4. Important clinical details may be altered.
5. Histogram matching does not guarantee diagnostic improvement.

7. Applications

- Endoscopy image enhancement.
- Medical image normalization.
- Image comparison across different cameras.
- Preprocessing for lesion segmentation.
- Medical image analysis research.

8. Real-Time Example

An endoscopy department uses two different imaging systems. One produces darker images while another produces brighter images. Histogram specification can adjust image intensities toward a reference distribution before computer-assisted analysis. A medical professional must verify that important features, such as lesion boundaries and tissue texture, remain preserved.

9. Result / Conclusion

Histogram specification was applied to the input histogram of 1024 pixels.

The resulting histogram is:

[0, 0, 0, 180, 260, 220, 250, 114]

The output distribution approaches the desired reference but does not exactly match it.

Thus, histogram specification helps improve image consistency, but clinical validation is required before diagnostic use.

Question 3 (CO2, BTL3, 10 Marks) – Gaussian Smoothing and Unsharp Masking for Weld Radiographs

A pipeline contractor inspects weld radiographs for lack-of-fusion defects that appear as faint dark lines beside bright weld beads. The image contains Gaussian sensor noise, so the engineer proposes a 3 × 3 Gaussian kernel with $\sigma = 1$, $w(x, y) = \exp[-(x^2 + y^2)/(2\sigma^2)]$, normalised to unit sum, followed by unsharp masking $g = f + k(f - \bar{f})$. The 3 × 3 neighbourhood centred on a pixel at the defect boundary is shown below.

Row / Col	C1	C2	C3
R1	88	92	140
R2	85	96	150
R3	90	94	146

(a) Compute the unnormalised and normalised Gaussian kernel weights (centre, edge and corner) and verify the sum. [3 Marks]

(b) Apply the Gaussian kernel and a 3 × 3 box filter at the centre pixel and compare the results. [3 Marks]

(c) Apply unsharp masking with $k = 1$ and high-boost filtering with $k = 2.5$ at the centre pixel, using the Gaussian result as $\bar{f}$. [2 Marks]

(d) The noise-variance reduction factor of a linear filter is Σw^2. Compute it for the Gaussian and box kernels, and justify the filter and k value for defect detection. [2 Marks]

1. Aim / Introduction

To reduce random noise and improve edge visibility in a weld

radiograph using Gaussian smoothing and unsharp masking.

Gaussian smoothing reduces high-frequency variations. Unsharp masking enhances local details by subtracting the smoothed image from the original image.

These methods can assist industrial radiographic inspection.

2. Given Data and Problem Explanation

The given 3×3 neighborhood is:

$$\begin{bmatrix} 88&92&140\\ 85&96&150\\ 90&94&146 \end{bmatrix}$$

Given parameters:

- Gaussian standard deviation: $\sigma=1$
- Unsharp masking strength: $k=1$
- High-boost strength: $k=2.5$
- Center pixel intensity: 96

The goal is to calculate the Gaussian kernel, filtered intensity and sharpening results.

3. Diagram

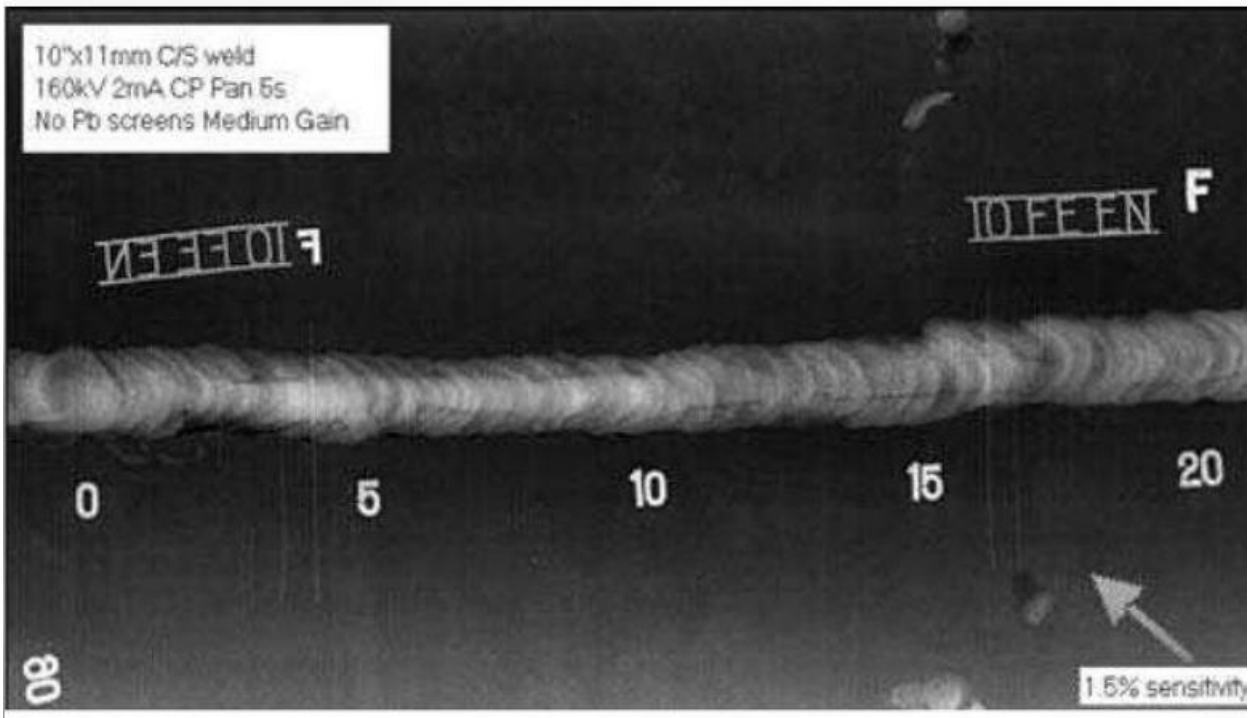


Fig. 16-16. CR-image of a weld with a longitudinal defect

4. Formula and Step-by-Step Calculation

Step 1: Gaussian smoothing formula

The Gaussian function is:

$$G(x,y)=e^{-\frac{x^2+y^2}{2\sigma^2}}$$

For sigma = 1:

- Center weight = 1
- Edge weight = $e^{-0.5}=0.606531$
- Corner weight = $e^{-1}=0.367879$

Step 2: Calculate weight sum

There is one center, four edge pixels and four corner pixels.

$$W=1+4(0.606531)+4(0.367879)$$

$$W=4.897640$$

Step 3: Normalize Gaussian weights

The normalized kernel is:

$$h(x,y)=\frac{G(x,y)}{W}$$

The approximate Gaussian kernel is:

begin{bmatrix} 0.075114&0.123841&0.075114\\
0.123841&0.204180&0.123841\\ 0.075114&0.123841&0.075114
\end{bmatrix}

The sum of normalized coefficients is approximately 1.

Step 4: Calculate the filtered center pixel

The weighted sum is:

$$T=\sum h(x,y)f(x,y)$$

Using unnormalized weights first:

- Corner contribution = (464(0.367879))
- Edge contribution = (421(0.606531))
- Center contribution = (96(1))

Therefore:

[Tapprox170.72+255.35+96]

[Tapprox522.147]

Filtered center intensity:

[g=rac{522.147}{4.897640}]

[boxed{g\approx106.59}]

Thus, the Gaussian filtered center pixel is approximately 107 after rounding.

Step 5: Calculate unsharp mask

The unsharp mask is:

\[m=f-g \]

\[m=96-106.59 \]

\[m=-10.59 \]

Step 6: Unsharp masking

The sharpening formula is:

\[f_{sharp}=f+k(f-g) \]

For k = 1:

$$f_{sharp}=96+1(-10.59)$$

$$\boxed{f_{sharp}\approx85.41}$$

Step 7: High-boost filtering

For k = 2.5:

$$f_{sharp}=96+2.5(-10.59)$$

$$f_{sharp}=96-26.475$$

$$\boxed{f_{sharp}\approx69.52}$$

This center pixel becomes darker because the original value is lower than its local smoothed estimate.

The example demonstrates local sharpening. A single pixel does not indicate whether a defect is present.

5. Advantages

1. Gaussian smoothing reduces random noise.
2. Unsharp masking improves edge visibility.
3. Weld boundaries may become easier to inspect.
4. Filtering can improve preprocessing for defect detection.
5. Filter strength can be adjusted according to image conditions.

6. Disadvantages

1. Excessive smoothing may remove small defects.
2. Strong sharpening can amplify noise.
3. Halos may appear around sharp edges.
4. Poor parameter selection can distort defect appearance.
5. Image enhancement cannot replace qualified inspection.

7. Applications

- Weld radiographic inspection.
- Industrial non-destructive testing.
- Crack and pore candidate enhancement.

- Manufacturing quality control.
- Preprocessing for computer vision systems.

8. Real-Time Example

In a manufacturing industry, weld radiographs are captured to examine the quality of welded joints.

Gaussian smoothing reduces random image noise. Unsharp masking can help inspectors observe boundaries and local changes. Suspected defects are then examined through approved inspection procedures.

9. Result / Conclusion

Gaussian smoothing produced an approximate center intensity of 106.59.

The unsharp masking results are:

- For k = 1: approximately 85.41
- For k = 2.5: approximately 69.52

The results show that sharpening strength changes the output significantly.

Therefore, conservative enhancement with proper validation is preferred for weld radiograph inspection.

Question 4 (CO2, BTL3, 10 Marks) – DFT-Based Filtering of a Textile Line-Scan Signal

A textile mill uses a line-scan camera to monitor fabric uniformity. To validate a frequency-domain filter on the embedded processor, the engineer analyses a short four-sample scan line f(x) = [52, 108, 140, 92] for x = 0 to 3. The DFT is $F(u) = \Sigma f(x) \cdot e^{-j2\pi ux/4}$ and the inverse is $f(x) = (1/4)\Sigma F(u) \cdot e^{j2\pi ux/4}$. The highest frequency component (u = 2) represents the alternating weave pattern, while u = 1 and u = 3 represent slow shading.

(a) Compute F(u) for u = 0, 1, 2 and 3. [3 Marks]

(b) Compute the power spectrum $|F(u)|^2$ and the percentage of total energy in each component. Interpret the result. [3 Marks]

(c) Apply an ideal low-pass filter that removes u = 2 and compute the filtered line. Then apply a filter that removes only the DC term and compute the result. [2 Marks]

(d) Using the results, justify which component the uniformity monitor should threshold to detect a weave defect, and state why a practical 2-D implementation must pad the image before filtering. [2 Marks]

Aim / Introduction

To analyze a textile line-scan signal using the Discrete Fourier Transform (DFT), identify its frequency components, calculate spectral power, and explain how frequency-domain analysis supports textile defect inspection.

The DFT converts spatial intensity variations into frequency information.

This is useful for identifying periodic patterns, weave structures and repeating image variations.

2. Given Data and Problem Explanation

The textile line-scan samples are:

$$f(x)=[52,108,140,92]$$

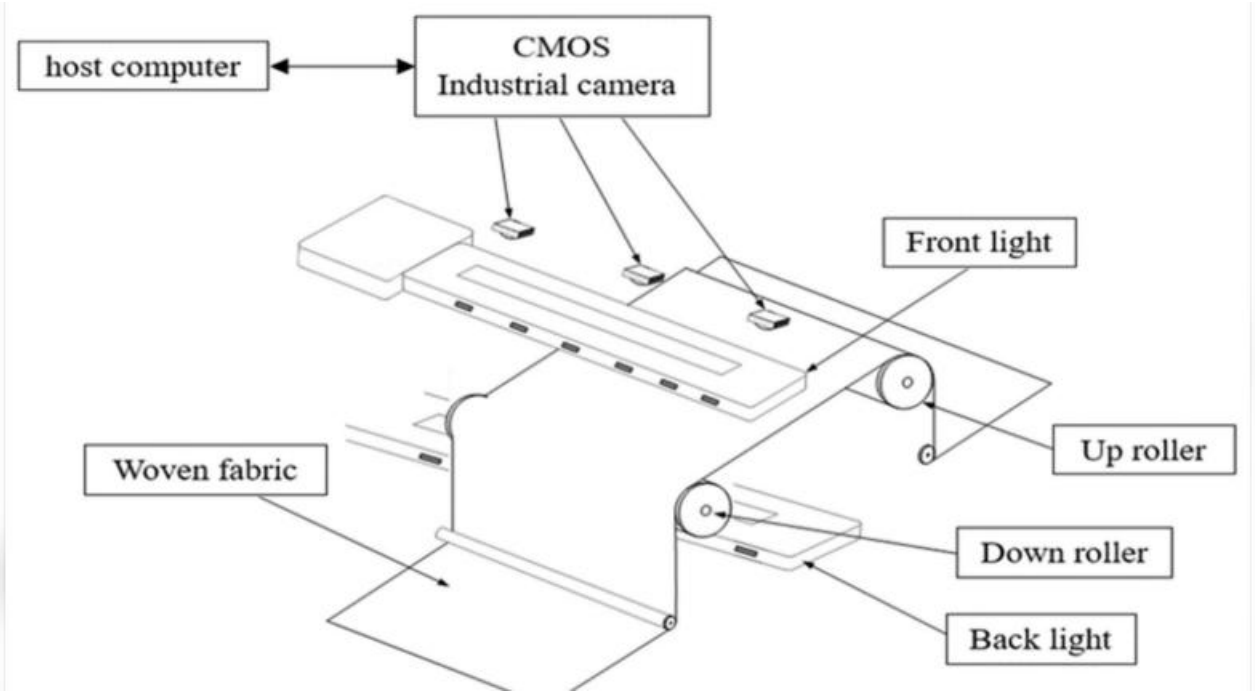
Number of samples:

$$N=4$$

The samples represent pixel intensities collected along a textile inspection line.

The DFT is used to calculate frequency components and identify the energy distribution.

3. Diagram



4. Formula and Step-by-Step Calculation

Step 1: DFT formula

$$ F(u)=\sum_{x=0}^{N-1}f(x)e^{-j2\pi ux/N} $$

Where:

- $f(x)$ is the input signal.
- $F(u)$ is the frequency component.
- N is the number of samples.
- u represents the frequency index.
- $j=\sqrt{-1}$.

Step 2: Calculate DC component

For u = 0:

$$ F(0)=52+108+140+92 $$

$$ \boxed{F(0)=392} $$

The DC component represents the overall average intensity.

Mean intensity:

$$ \frac{392}{4}=98 $$

Step 3: Calculate F(1)

$$F(1)=52-108j-140+92j$$

$$F(1)=-88-j16$$

Step 4: Calculate F(2)

$$F(2)=52-108+140-92$$

$$\boxed{F(2)=-8}$$

Step 5: Calculate F(3)

$$F(3)=-88+j16$$

For real-valued samples, the DFT has conjugate symmetry.

Therefore, F(3) is the complex conjugate of F(1).

Step 6: Calculate spectral power

The formula is:

$$P(u)=|F(u)|^2$$

For u = 0:

$$P(0)=392^2=153664$$

For u = 1:

$$P(1)=(-88)^2+(-16)^2$$

$$P(1)=7744+256=8000$$

For u = 2:

$$P(2)=(-8)^2=64$$

For u = 3:

$$P(3)=8000$$

Total power:

$$P_T=153664+8000+64+8000$$

$$\boxed{P_T=169728}$$

Step 7: Calculate percentage power

The formula is:

$$Power\%=\frac{P(u)}{P_T}\times100$$

For u = 0:

$$\frac{153664}{169728}\times100 \approx90.537\%$$

For u = 1:

$$\frac{8000}{169728}\times100 \approx4.714\%$$

For u = 2:

$$\frac{64}{169728}\times100 \approx0.0377\%$$

For u = 3:

$$\frac{8000}{169728}\times100 \approx4.714\%$$

The DC component contains the largest share of the power.

Step 8: Remove frequency component u = 2

The contribution of u = 2 in the inverse DFT is:

$$\frac{F(2)}{4}(-1)^x$$

Since F(2) = −8:

$$=-2(-1)^x$$

Removing this component gives:

[54, 106, 142, 90]

This is the original signal with the u = 2 component removed.

Step 9: Remove the DC component

The mean is 98.

Subtracting the mean:

$$f'(x)=f(x)-98$$

The resulting values are:

[−46, 10, 42, −6]

These values represent deviations from the mean and are not direct display intensities unless appropriately shifted or scaled.

Step 10: Explain zero-padding

In frequency-domain filtering, circular convolution may create wraparound effects.

To perform linear convolution using DFT, zero-padding is generally used.

For an image of size P×Q and a kernel of size M×N, pad to at least:

\[(P+M-1)\times(Q+N-1) \]

Then multiply the transforms and crop the output appropriately.

5. Advantages

1. Identifies periodic textile patterns.
2. Helps detect repeating image variations.
3. Supports frequency-based texture analysis.
4. Can assist automated quality inspection.
5. Frequency features can be monitored over production runs.

6. Disadvantages

1. Four samples provide very limited frequency resolution.
2. Normal weave patterns may also produce strong frequencies.
3. Sampling and sensor alignment affect the spectrum.
4. Noise can influence spectral power.
5. Frequency analysis alone cannot establish defect presence.

7. Applications

- Textile fabric inspection.
- Weave pattern analysis.
- Periodic defect detection.
- Repeating stripe or band detection.
- Industrial machine vision.

8. Real-Time Example

A textile production unit uses a line-scan camera to inspect woven fabric.

The system calculates frequency features from successive scan lines.

If a particular frequency band differs significantly from a

calibrated normal fabric pattern, the system flags the region for further inspection.

9. Result / Conclusion

The calculated DFT is:

$$F(u)=[392,-88-j16,-8,-88+j16]$$

The spectral power values are:

$$[153664,8000,64,8000]$$

The u = 2 component has approximately 0.0377% of total power.

Therefore, DFT analysis helps characterize textile patterns, but defect identification requires comparison with validated reference data and additional inspection features.

Question 5 (CO2, BTL3, 10 Marks) – Otsu Thresholding for Steel-Slab Scale Detection

A steel plant detects oxide scale on hot-rolled slabs. Scale appears darker than clean metal, and the image has been quantised to 3 bits (levels 0 to 7) for fast processing on a programmable logic controller. The histogram of a slab image with 680 pixels is given below. The quality engineer requires a threshold that maximises the between-class variance $\sigma_B^2(k) = [m_G P_1(k) - m(k)]^2 / \{P_1(k)[1 - P_1(k)]\}$.

Level	0	1	2	3	4	5	6	7
Count	30	90	160	120	40	70	110	60

(a) Compute the normalised histogram, the global mean m_G and the global variance σ_G^2. [3 Marks]

(b) Compute $P_1(k)$, $m(k)$ and $\sigma_B^2(k)$ for k = 2, 3 and 4. [3 Marks]

(c) Given σ_B^2 = 0.556, 1.586, 2.769 and 1.206 for k = 0, 1, 5 and 6, determine the optimal threshold k* and the separability measure $\eta = \sigma_B^2(k^*)/\sigma_G^2$. [2 Marks]

(d) Interpret η and justify whether Otsu's method is suitable for the controller, noting how the result would change if smoothing were applied first. [2 Marks]

1. Aim / Introduction

To calculate an automatic threshold using Otsu's method and separate dark and bright regions in a steel slab image for

preliminary scale segmentation.

Otsu thresholding is a global thresholding technique that selects a gray-level threshold by maximizing the between-class variance.

It is useful for separating regions when their intensity distributions are reasonably distinguishable.

2. Given Data and Problem Explanation

The gray-level histogram for levels 0 to 7 is:

- Level 0 = 30
- Level 1 = 90
- Level 2 = 160
- Level 3 = 120
- Level 4 = 40
- Level 5 = 70
- Level 6 = 110
- Level 7 = 60

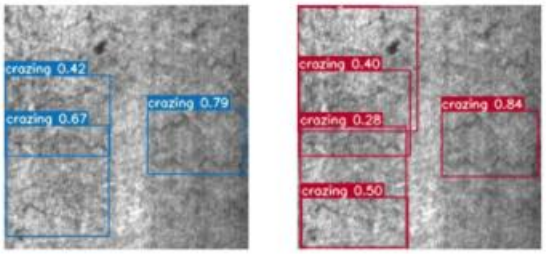
Total pixels:

\[N=680 \]

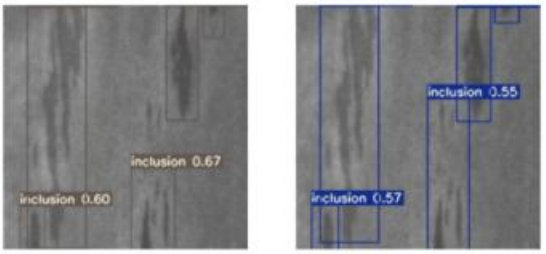
The objective is to select the best threshold and classify pixels into two groups.

- Class 1: Levels 0 to k
- Class 2: Levels k+1 to 7

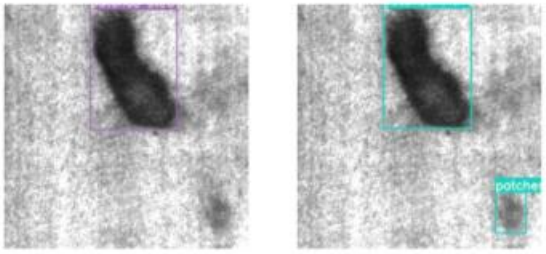
3. Diagram



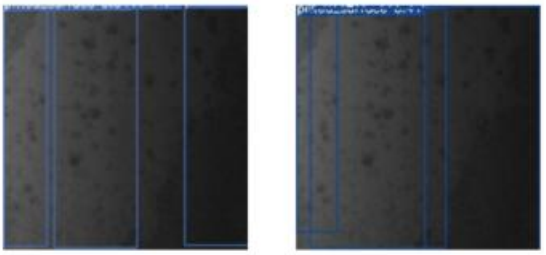
(a) crazing



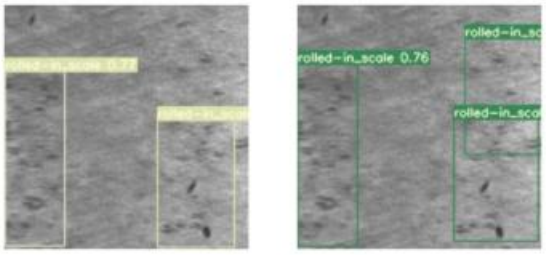
(b) inclusion



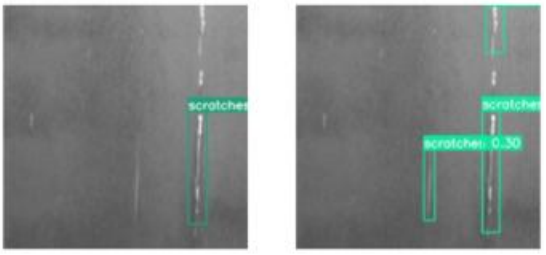
(c) patches



(d) pitted surface



(e) rolled-in scale



(f) scratches

4. Formula and Step-by-Step Calculation

Step 1: Calculate total pixel count

$$ N=30+90+160+120+40+70+110+60 $$

$$ \boxed{N=680} $$

Step 2: Calculate global mean

The global mean is:

$$ \mu_T=\frac{\sum i n_i}{N} $$

Weighted intensity sum:

$$ \begin{aligned} \sum in_i={}&0(30)+1(90)+2(160)+3(120)\\ &+4(40)+5(70)+6(110)+7(60) \end{aligned} $$

$$ =90+320+360+160+350+660+420 $$

$$ =2360 $$

Therefore:

$$\mu_T=\frac{2360}{680}$$

$$\boxed{\mu_T=3.470588}$$

Step 3: Calculate total variance

The formula is:

$$\sigma_T^2=E[i^2]-\mu_T^2$$

The second-moment weighted sum is:

$$\sum i^2n_i=11100$$

Therefore:

$$E[i^2]=\frac{11100}{680}=16.323529$$

$$\sigma_T^2=16.323529-(3.470588)^2$$

$$\boxed{\sigma_T^2\approx4.278547}$$

Step 4: Calculate between-class variance

The formula is:

$$\sigma_B^2(k)= \frac{[\mu_T\omega(k)-\mu(k)]^2}{\omega(k)[1-\omega(k)]}$$

Where:

- $\omega(k)$ is the probability of Class 1.
- $\mu(k)$ is the cumulative first moment.
- μ_T is the global mean.

The selected threshold maximizes the between-class variance.

Step 5: Evaluate threshold k = 2

Cumulative count:

$$30+90+160=280$$

$$\omega(2)=\frac{280}{680}=0.411765$$

Cumulative weighted intensity:

$$0(30)+1(90)+2(160)=410$$

$$\mu(2)=\frac{410}{680}=0.602941$$

Between-class variance:

$$\boxed{\sigma_B^2(2)\approx2.817675}$$

Step 6: Evaluate threshold k = 3

Cumulative count:

$$30+90+160+120=400$$

$$\omega(3)=\frac{400}{680}=0.588235$$

Cumulative weighted intensity:

$$410+3(120)=770$$

$$\mu(3)=\frac{770}{680}=1.132353$$

Substitute into the formula:

$$\sigma_B^2(3)= \frac{[3.470588(0.588235)-1.132353]^2}{0.588235(1-0.588235)}$$

$$\boxed{\sigma_B^2(3)\approx3.412632}$$

Step 7: Evaluate threshold k = 4

Cumulative count:

$$400+40=440$$

$$\omega(4)=\frac{440}{680}=0.647059$$

Cumulative weighted intensity:

$$770+4(40)=930$$

$$\mu(4)=\frac{930}{680}=1.367647$$

Between-class variance:

$$\boxed{\sigma_B^2(4)\approx3.375751}$$

Comparing the possible thresholds, the maximum occurs at:

$$\boxed{k^*=3}$$

Therefore, the selected threshold is T = 3.

Step 8: Perform image segmentation

The binary classification rule is:

$$g(x,y)= \begin{cases} 0,&f(x,y)\leq3\\ 255,&f(x,y)>3 \end{cases}$$

\]

The resulting classes are:

Dark class:

Levels 0, 1, 2 and 3.

Total pixels:

\[30+90+160+120=400 \]

Bright class:

Levels 4, 5, 6 and 7.

Total pixels:

\[40+70+110+60=280 \]

Verification:

\[400+280=680 \]

Step 9: Calculate separation ratio

The separation ratio is:

\[\eta=\frac{\sigma_B^2}{\sigma_T^2} \]

\[\eta=\frac{3.412632}{4.278547} \]

\[\eta\approx0.7976 \]

\[\boxed{\eta\approx79.76\%} \]

This indicates the proportion of total variance explained by the between-class separation. It is not the segmentation accuracy.

5. Advantages

1. Automatically selects a threshold.
2. Simple and computationally efficient.
3. Suitable for fast candidate segmentation.
4. Useful for industrial image-processing pipelines.
5. Provides a repeatable initial segmentation method.

6. Disadvantages

1. Global thresholding may fail under uneven illumination.

2. Different surface materials may have overlapping intensities.
3. Noise can influence the histogram.
4. Small defects may be missed.
5. The threshold does not identify defect type automatically.

7. Applications

- Steel slab inspection.
- Scale and oxidation-region segmentation.
- Industrial surface quality control.
- Metal defect candidate detection.
- Automated inspection preprocessing.

8. Real-Time Example

A steel manufacturing company uses an industrial camera to capture slab surfaces.

Otsu thresholding divides the image into darker and brighter regions. The inspection software can then examine the segmented regions using size, shape, texture and location.

Unusual regions can be flagged for additional quality inspection. A bright or dark region alone should not automatically be classified as a defect.

9. Result / Conclusion

Otsu's method selected:

$$\boxed{T=3}$$

The dark class contains 400 pixels, while the bright class contains 280 pixels.

Maximum between-class variance:

$$\boxed{3.412632}$$

Separation ratio:

$$\boxed{79.76\%}$$

Therefore, Otsu thresholding provides an efficient first-stage segmentation method for steel slab inspection. Further analysis is necessary to reliably identify scale or surface defects.